Biswajit Pain

Systems Development Engineer / Platform Engineer

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PROFILE

Systems Development Engineer with 12+ years building and operating production cloud and distributed-storage infrastructure across AWS, Delivery Hero, Sixt, and VMware. Hands-on experience with Ceph (RBD, RGW, CephFS), Rook on Kubernetes, OpenStack (Cinder, Swift, Nova), and VMware vSAN / vSphere storage, delivered in both production and internal-platform environments. Day-to-day engineering in Go, Python, and Bash across AWS, GCP, and private datacenters; deep troubleshooting through Linux, networking, and the storage layer - from block device and filesystem behaviour up to S3 / object gateway semantics.

PROFESSIONAL EXPERIENCE

Lead Systems Development Engineer
October 2022 - Present

Amazon Web Services (AWS) - Berlin, Germany

Database service - region build and storage layout

- Owned the redesign of the region build process for a managed database service with a large distributed storage footprint, deployed across AWS regions.
- Rebuilt the provisioning pipeline so new regions come up with correct control-plane topology, storage layout, replication configuration, and networking without manual intervention.
- Shortened time-to-first-customer in a new region and removed a class of bootstrap failures caused by race conditions in the storage and control-plane initialization flow.

Sovereign Cloud - Berlin engineering org

- Part of the founding engineering group setting up AWS European Sovereign Cloud infrastructure in Berlin.
- Ran hundreds of technical interviews (systems design, coding, service deep-dive) to build the local engineering bench for a regulated, isolated cloud region.
- Contributed to early technical decisions on service isolation, operator access, and data-residency boundaries.

Game orchestration platform

- Engineer on the platform that orchestrates game server fleets for one of the largest game-hosting services on AWS.
- Worked on scheduling, capacity, and low-latency session placement across regions.

Team & operations

- Technical lead for 10 engineers split across Berlin and a second continent; own architecture reviews, design docs, and on-call quality for the team's services.

Staff Systems Engineer
April 2020 - September 2022

Delivery Hero SE - Berlin, Germany

Zero-downtime MySQL migration on GCP

- Led migration of a large production MySQL estate to Google Cloud SQL for a globally used ordering platform.
- Designed the cutover using replication, read-only windows, and application-level dual-write verification so there was no visible downtime for restaurants or customers.
- Built pre- and post-migration consistency checks (row counts, checksum-based diff on critical tables) to prove correctness before committing to the new primary.

Fintech payment reliability

- Debugged and fixed recurring failures in the payment processing path: idempotency gaps, duplicate-charge edge cases, and timeouts against external PSPs.
- Added retry/backoff, circuit breakers, and clearer reconciliation jobs; cut payment-related incident volume and improved settlement accuracy.

Internal tooling

- Wrote internal CLIs and dashboards (Python + Grafana) to speed up incident triage across the distributed payment services.

Manager III, Site Reliability Engineering
April 2018 - March 2020

Sixt Research & Development - Berlin / Munich, Germany

Hydrant - infrastructure automation platform

- Led Hydrant - an internal platform that automated provisioning of AWS accounts, VPCs, IAM baselines, and standard service stacks for product teams.
- Replaced hand-rolled tickets and ad-hoc Terraform usage with a self-service flow; cut new-environment setup from days to

under an hour.
- Stack: Terraform, Python, Jenkins, AWS Organizations.

PCI DSS-compliant payment platform on AWS
- Designed and deployed the payment platform against PCI DSS requirements: network segmentation, restricted IAM, centralized logging, encryption at rest and in transit, audit trails.
- Drove the audit with external assessors and passed compliance on first attempt.

Team leadership
- Managed 7 SREs across Munich and Bangalore; ran hiring, on-call rotations, and roadmap planning across two sites.

Member of Technical Staff II — November 2016 - March 2018

VMware Software India Pvt. Ltd. - Bangalore, India

Ceph storage for multi-cloud IaaS
- Designed and operated Ceph clusters providing block (RBD), object (RGW / S3-compatible), and file (CephFS) services backing the multi-cloud IaaS platform.
- Sized and built the CRUSH map and pool layout (replicated and erasure-coded pools, separate device classes for SSD and HDD tiers) to match durability and latency targets.
- Tuned BlueStore OSDs, placement group counts, scrub schedules, and recovery/backfill throttles to keep client IO predictable during node loss and rebalance events.
- Operated RGW for S3-compatible object storage: bucket lifecycle policies, user and quota management, and multi-site replication between sites.

Rook operator on Kubernetes
- Deployed and operated Rook to run Ceph on Kubernetes for internal platforms, automating cluster bring-up, OSD lifecycle on node churn, and version upgrades.
- Built StorageClasses and CSI bindings so product teams could consume RBD volumes and CephFS shares as first-class Kubernetes PersistentVolumes.
- Wrote runbooks and automation around the Rook CRDs for day-2 operations: capacity expansion, pool changes, and failure recovery.

OpenStack + Ceph integration
- Integrated Ceph with OpenStack services - RBD as the backend for Cinder (block) and Glance (image), Swift-compatible object storage via RGW.
- Worked across Nova, Cinder, Neutron, and Keystone to debug cross-component issues (volume attach races, auth token failures, network policy and QoS at the storage network layer).
- Reviewed tenant isolation, rate-limiting, and quota enforcement for multi-tenant shared storage.

Platform engineering - microservices and CI
- Broke out services from a large monolithic test/CI system into independently deployable microservices with clear API boundaries; introduced service-level health checks and per-service deployment pipelines.

Production Engineer — January 2016 - October 2016

ANI Technologies (Ola) - Bangalore, India

HAProxy log analytics pipeline
- Built the HAProxy log pipeline using rsyslog, Kafka, Logstash, and Elasticsearch for real-time visibility into traffic patterns and error rates across the fleet.
- Used the pipeline to identify slow upstreams and misbehaving clients that had previously been invisible.

Microservice infrastructure
- Contributed to the platform supporting the company's rapid user growth: service discovery, deploy tooling, and base images.

DevOps Engineer — April 2014 - December 2015

Knowlarity - Bangalore, India

Multi-datacenter VoIP telephony infrastructure
- Designed and built the multi-datacenter VoIP telephony infrastructure powering the company's cloud telephony product across India and SEA.
- Built the active-active datacenter layout with SIP trunk termination across sites, media server pools, and carrier-level redundancy so call traffic survived single-DC loss.
- Owned the failover platform end-to-end: DNS, load balancer, Asterisk/FreeSWITCH media nodes, and DB replica cutover runbooks.

Hub-and-spoke network
- Designed and rolled out a hub-and-spoke network connecting multiple datacenters, office sites, and carrier interconnects with redundant IPsec tunnels and dynamic routing.
- Replaced an unstable full-mesh VPN setup; resolved persistent inter-site routing loops, asymmetric paths, and fragmentation

  issues that had been causing dropped calls and signaling failures.
- Tuned OSPF/BGP route preferences and failover timers so spoke sites re-converged on the surviving hub within seconds of a link loss.

Assistant Systems Engineer / Trainee July 2013 - March 2014

Tata Consultancy Services (TCS) - Kolkata, India

- Delivered infrastructure automation work including storage virtualization tasks for a client engagement.
- Led a team of five during the Initial Learning Program.

TECHNICAL SKILLS

Languages:	Go, Python, Bash, C, Java
Kubernetes & Containers:	Kubernetes, Helm, container runtimes, CRDs / operators, microservices
Cloud & IaaS:	AWS (EC2, S3, Route53, RDS, CloudFront, CloudWatch, SES, SNS, IAM), GCP, VMware IaaS
Ceph & Distributed Storage:	Ceph RBD, RGW, CephFS; Rook on Kubernetes; CRUSH maps, placement groups, BlueStore OSDs, erasure coding, replication factors, pool and bucket design; cluster recovery, scrub / backfill tuning, RGW multi-site
OpenStack:	Cinder (block), Swift (object), Nova (compute), Keystone, Neutron; Ceph-backed Cinder and Glance
Storage:	VMware vSAN, vSphere storage, enterprise storage (IBM, Hitachi), NAS, S3 / object storage, block volumes, replication, failover
Data:	MySQL, PostgreSQL, Kafka, Elasticsearch, data migration
Infra as Code / CI/CD:	Terraform, Chef, Puppet, Jenkins, Git-based pipelines
Observability:	OpenTSDB, Graylog, Fluentd, Logstash, rsyslog, cAdvisor, Nagios
OS / Networking:	Linux (RHEL, Ubuntu), performance tuning, TCP/IP, VPN (hub-and-spoke), HAProxy, load balancing
Web Frameworks:	Django, Flask

EDUCATION

B.Tech, Computer Science and Engineering

Kalyani Government Engineering College, India (2013)